Synthetic algebraic geometry in Unit: what has been formalised

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Abstract

We report on a first formalisation of parts of *A foundation for synthetic algebraic geometry* (Cherubini, Coquand, Hutzler [1]) in the prototype Unit, the version where recursion is only through eliminators. The foundation is a univalent universe with Rijke's axiom of replacement, as formulated in [3]; propositional truncation and set quotients are derived from it. Finitely presented algebras are given by presentations and a universal property, and every presentation has a presented algebra. SQC is one uniform hypothesis on all finitely presented algebras. The results checked are Proposition 2.2.2, Lemma 2.2.4 (from Loc and SQC only), Theorem 3.3.3, Example 6.1.9 (the fibres computed in the example, and the coordinate algebra of a fibre), Lemma 6.2.1, Lemma 7.4.2, Theorem 7.4.7 and Theorem 7.4.8. Each development checks in less than half a second.

1 The foundation

All developments share one segment whose parameters are the foundational axioms:

- *univalence*, at every level, in the form "idtoeqv is an equivalence";
- *replacement*, as two parameters which together form one axiom schema [3, Definitions 2, 4, 6]:
  - `images`: for $A : \mathsf{U}_l$, a locally U_l-small $B : \mathsf{U}_k$ and $f : A \to B$, a chosen factorisation $A \to I \to B$ with $I : \mathsf{U}_l$, a commuting triangle, and $I \to B$ an embedding;
  - `surjections`: the map $A \to I$ is surjective in the sense that precomposition is an equivalence on sections of every family of propositions over I, at every level j.

The chosen image does not depend on the level j of elimination. The schema is split into data and a property only because a type quantifying over all levels does not belong to a universe. No resizing is assumed.

From these parameters the following are derived, not assumed:

- function extensionality, from univalence;
- the propositional truncation $\|A\|$, as the image of $A \to 1$, with induction into propositions of any level and its computation rule as an identity;
- set quotients of proposition-valued equivalence relations, as the image of $a \mapsto (x \mapsto R(x, a))$ into $A \to \mathsf{Prop}$ (locally small by univalence); they are effective, and maps into sets that respect the relation descend to them;
- univalent completions of small families.

The computation rules of the derived truncation and quotients are proved identities. The evaluator has no new reduction rules.

2 How the mathematics is represented

Rings. A commutative ring is a record with a carrier, the operations, and the axioms as identities. Where needed the carrier is assumed to be a set. A unit is an element with a chosen inverse, which is a proposition when the carrier is a set.

Polynomials. A polynomial is a list of coefficients, lowest degree first. Two lists are equal as polynomials when all their coefficients are equal, so trailing zeros do not matter. In the files on $\mathbb{P}^1$ the type $R[X]$ is the quotient of lists by this relation; in the file on Lemma 2.2.4 we keep lists and the relation, which suffices. Evaluation is Horner's rule, a primitive recursion. Laurent polynomials are pairs of lists (p, n), standing for $p(X) + X^{-1}n(X^{-1})$.

Recursion. All recursive definitions and all proofs by induction are written with the eliminators of the data types, as in system T but with dependent types. Several proofs use recursion at higher type. For instance, the induction on the degree in Lemma 2.2.4 returns, at each step, a function over all polynomials of the smaller length.

Finitely presented algebras. An R-algebra is a commutative ring whose carrier is a set, with a ring homomorphism from R. A presentation is a number n and a list of expressions in the variables $\mathsf{Fin}\, n$; expressions are an ordinary data type (variables, constants of R, sums, products), and Fin is defined by recursion ($\mathsf{Fin}\, 0 = \emptyset$, $\mathsf{Fin}(n+1) = \mathsf{Fin}\, n + 1$), so no indexed families are needed. The solutions of P in an algebra B are the assignments $V \to B$ on which every equation evaluates to 0. An algebra A is *presented* by P when it has a solution a of P such that for every algebra B the map $\mathrm{Hom}(A, B) \to \mathrm{Sol}_P(B)$, $\varphi \mapsto \varphi \circ a$, is an equivalence; A is *finitely presented* when merely there is a presentation of it, as in [1, Definition 1.3.2]. Then $\mathrm{Spec}A = \mathrm{Hom}(A, R)$ is equivalent to $\mathrm{Sol}_P(R)$, the solutions of the equations in R. Every presentation has a presented algebra: expressions modulo the relation of merely having a finite derivation from the ring laws and the equations. A derivation is an ordinary tree with a validity predicate defined by recursion. Two algebras with the same presentation are isomorphic. The polynomial ring $A[X]$, built as a quotient of lists, has the universal property $\mathrm{Hom}(A[X], B) \simeq \mathrm{Hom}(A, B) \times B$, and is presented by any presentation of A with one more variable.

The axioms of [1]. Loc, SQC and Z-choice are not part of the foundation. When a proof needs one of them, it is a hypothesis of the segment containing the proof. SQC is stated once: for every finitely presented algebra A, evaluation $A \to (\mathrm{Spec}A \to R)$ is an equivalence. Some earlier files use instances of SQC for particular algebras instead; this is said for each result. Z-choice is not used.

3 The results

Proposition 2.2.2: the weak Nullstellensatz

For a finitely presented A, $\mathrm{Spec}A = \emptyset$ if and only if $1 = 0$ in A. By SQC, an empty spectrum gives $A \simeq (\emptyset \to R)$, a one-element ring; the converse uses $1 \neq 0$ in R.

For $A = R[X]/(f)$, presented by one variable and the equation f, the algebraic content is a certificate: if $1 = 0$ in A then merely $fg = 1$ for some list g, with equality of coefficients. It is obtained by reading a finite derivation of $1 = 0$ as a computation of a multiplier: each rule (the ring laws, the equation $f = 0$, symmetry, transitivity, compatibility with $+$ and $\times$) transforms multipliers r with $x = y + fr$. No second quotient of $R[X]$ by an ideal is built, and SQC is not used for this step. Combined with SQC: if f has no root then merely $fg = 1$ for some g.

Lemma 2.2.4: R is algebraically closed up to double negation

For a polynomial f, it is not not the case that $f = 0$ or

$$f = \alpha\,(X - a_1)^{e_1} \cdots (X - a_n)^{e_n},$$

with α a unit, $e_i \geq 1$ and the a_i pairwise distinct.

The proof follows the paper: induction on a bound for the degree, inside the double negation monad. The top coefficient is zero (drop it) or not. In the second case it is a unit, and if the degree is at least 1 there is not not a root a. Then synthetic division by $X - a$ gives $f = (X - a)q$, and q has a smaller bound. Synthetic division is one primitive recursion returning both the quotient and the remainder, and gives the factor theorem with the bound on the degree. A last induction groups equal roots, deciding $a_i = a_j$ under the double negation. This uses that $X - a$ and $X - b$ commute, a coefficient computation.

Hypotheses: Loc and SQC. The proof was first written with two hypotheses, $1 \neq 0$ and the weak Nullstellensatz for $R[X]/(f)$ in the form: if f has no root then it is not not the case that $fg = 1$ for some list g. Double negation suffices because the proof only uses it inside double negations, and the mere inverse given by Proposition 2.2.2 implies it. With Proposition 2.2.2 the lemma follows from Loc and SQC alone. From the Nullstellensatz we prove Proposition 2.2.3(a) up to double negation (a constant $x \neq 0$ has no root, hence is not not invertible). We also prove that a polynomial of degree ≥ 1 with a unit top coefficient is not invertible: the top coefficient of fg is the product of the top coefficients. Replacement is not needed here. Nine computations on $\mathbb{Z}/2$ check that the definitions compute, for instance $1 + X^2 = (X + 1)^2$.

Theorem 3.3.3: functions $\mathrm{Spec}A \to \mathbb{N}$ are bounded

For a finitely presented A and $f : \mathrm{Spec}A \to \mathbb{N}$, merely there is N with $f(x) \leq N$ for all x. The paper's proof goes through tensor products (Lemma 3.3.1) and eventually vanishing sequences. A shorter route uses only $A[T]$:

- the function $H(x, t) = t^{f(x)}$ on $\mathrm{Spec}A \times R$ is a function on $\mathrm{Spec}(A[T])$, since a homomorphism $A[T] \to R$ is a point of $\mathrm{Spec}A$ and the image of T;
- by SQC for $A[T]$, H is merely the evaluation of one polynomial p over A;
- at a point x, the polynomial obtained by applying x to the coefficients of p has the same values as $T^{f(x)}$, hence the same coefficients (SQC for $R[T]$);
- if $f(x)$ is at least the length of p, its coefficient at $f(x)$ is 0, while that of $T^{f(x)}$ is 1; this contradicts $1 \neq 0$.

The bound is the length of p. It is a mere bound because p is merely a list. Order on $\mathbb{N}$ is decidable, so no excluded middle is used. There are two versions: one with $1 \neq 0$ and the two halves of SQC used (surjectivity for $A[T]$, injectivity for $R[T]$) as hypotheses, and one from $1 \neq 0$ and uniform SQC, where $A[T]$ and $R[T]$ are shown finitely presented.

Example 6.1.9: squaring on the projective line

$\mathbb{P}^1$ is the quotient of two copies of R (the charts $[a:1]$ and $[1:b]$) by the relation $ab = 1$. It is shown equivalent to the quotient of unimodular pairs by multiplication with units. Squaring descends to $s : \mathbb{P}^1 \to \mathbb{P}^1$, with $s([x:y]) = [x^2 : y^2]$. The two fibres are computed:

$$s^{-1}([0:1]) \simeq \{a : R \mid a^2 = 0\}, \qquad s^{-1}([1:1]) \simeq \mathsf{Bool} \quad \text{if } 2 \neq 0.$$

The first is the case $t = 0$ of a chart computation valid for every t, $s^{-1}([t:1]) \simeq \{a \mid a^2 = t\}$, which only needs a ring. The second uses Loc, $1 \neq 0$, and that 2 is invertible. That $2 \neq 0$ implies 2 invertible uses SQC for the algebras $R/(x, y)$, given as a quotient of R, with evaluation into functions on $\{x = 0 \wedge y = 0\}$. These are the algebras used in the proof of Proposition 2.2.3.

The coordinate algebra of a fibre is treated in the setting of presentations. Let K be the algebra presented by one variable T and the equation $T^2 = a$. Then $(b, c) \mapsto b + cT$ is an equivalence $R \times R \simeq K$ compatible with the module operations, so K is free of rank two with basis $1, T$. With SQC, the functions on $\{x : R \mid x^2 = a\}$ are exactly the $x \mapsto b + cx$. For $a = 0$ this is first-order expansion on the infinitesimal disk: every f satisfies $f(d) = f(0) + f' d$ for a unique slope f'. This uses neither Loc nor the invertibility of 2. The connection with the fibres of s computed above is not made formally.

Lemma 6.2.1: functions $\mathbb{P}^1 \to R$ are constant

The map $R \to (\mathbb{P}^1 \to R)$ sending c to the constant function is an equivalence, with inverse evaluation at $[0:1]$. The hypotheses are SQC for $R[X]$ (evaluation $R[X] \to (R \to R)$ is an equivalence) and SQC for $R[X, X^{-1}]$ (evaluation into functions on $\{(x, y) \mid xy = 1\}$). The two restrictions of f to the charts are merely polynomials $p(X)$ and $q(X^{-1})$. They agree on the overlap, so by injectivity for the Laurent polynomials all positive coefficients vanish and the constants agree. The coefficient computation does not use SQC. Constancy then follows by induction over the quotient $\mathbb{P}^1$.

Lemma 7.4.2: cocycles and torsors with local sections

Let A be an abelian group and L a type with $\|L\|$. An A-torsor is a set T with a map $A \times T \to T$ such that $(a + b) \cdot t = a \cdot (b \cdot t)$ and $0 \cdot t = t$; for all t, t' there is exactly one a with $a \cdot t = t'$, written $t' - t$; and $\|T\|$. An L-cocycle is $c : L \times L \to A$ with $c_{ij} + c_{jk} = c_{ik}$. The formal statement is an equivalence

$$\textstyle\sum_{T : A\text{-torsor}} (L \to T) \;\simeq\; L\text{-cocycles}.$$

No axiom of [1] is used. Here is the formal proof, written in the usual style.

From torsors to cocycles. Given (T, u) with $u : L \to T$, let $c_{ij} = u_i - u_j$. This is a cocycle, since $(c_{ij} + c_{jk}) \cdot u_k = c_{ij} \cdot (c_{jk} \cdot u_k) = c_{ij} \cdot u_j = u_i$, and $u_i - u_k$ is the only element with this property.

From cocycles to torsors. Given c, let

$$T_c = \{v : L \to A \mid v_m = v_k + c_{km} \text{ for all } k, m\}, \qquad (a \cdot v)_k = a + v_k, \qquad \hat{u}_i = (k \mapsto c_{ik}).$$

We have $\hat{u}_i \in T_c$ since $c_{im} = c_{ik} + c_{km}$. T_c is a torsor: it is a set, as a subset of A^L; the action laws hold pointwise. For $v, w \in T_c$, the existence and uniqueness of $w - v$ is a proposition, so we may assume $k_0 : L$. Then $a = w_{k_0} - v_{k_0}$ works: $a + v_m = (w_{k_0} - v_{k_0}) + (v_{k_0} + c_{k_0 m}) = w_{k_0} + c_{k_0 m} = w_m$. It is unique, since $a' + v_{k_0} = w_{k_0}$ forces $a' = a$. Finally, $\|L\|$ and $i \mapsto \hat{u}_i$ give $\|T_c\|$.

One composite. The cocycle of $(T_c, \hat{u})$ is c: we need $c_{ij} \cdot \hat{u}_j = \hat{u}_i$, that is $c_{ij} + c_{jk} = c_{ik}$ for all k.

The other composite. Let c be the cocycle of (T, u) and

$$e : T \to T_c, \qquad e(t) = (k \mapsto t - u_k).$$

Then $e(t) \in T_c$, since $t - u_m = (t - u_k) + (u_k - u_m)$; e is equivariant, since $(a \cdot t) - u_k = a + (t - u_k)$; and $e(u_i) = \hat{u}_i$ by definition. e is a bijection. It is a proposition that each fibre has exactly one element, so we may assume $k_0 : L$. The element $y_{k_0} \cdot u_{k_0}$ is mapped to y, and $e(t) = y$ gives $t = (t - u_{k_0}) \cdot u_{k_0} = y_{k_0} \cdot u_{k_0}$. By univalence, an equivariant bijection preserving the points is an equality $(T, u) = (T_c, \hat{u})$: by induction on equivalences it is enough to take e the identity. Then the two actions are equal by function extensionality, the torsor laws are propositions, and the points agree.

Comparison with the paper.

- The paper takes $T_c = \{u \mid u_i - u_j = c_{ij}\}$ with points $k \mapsto c_{ki}$. Its own computation gives the cocycle $\tilde{c}_{ij} = c_{ji}$, so (T_c, c) is a preimage of the transposed cocycle $-c$, not of c. The definitions above correct this.
- The paper does not check that T_c is a set with a free and transitive action. This is the second place where a point of L is assumed.
- The paper shows that the left side is a set, that the map is an embedding (comparing two arbitrary (T, u) and (T', u')), and that it is surjective. Its map $T \to T'$ depends on a chosen $i : L$. The formal proof gives an inverse instead, and compares each (T, u) only with $(T_c, \hat{u})$. The map e needs no chosen point, and the left side being a set is not needed.
- "$(T, u) = (T', u')$ is equivalent to an isomorphism" is one line in the paper. In the formal proof it is the only use of univalence, and only one direction is needed.
- The axiom $0 \cdot t = t$ is part of the definition, but no step of the proof uses it. The left side lies in a larger universe than the cocycles.

Appendix A shows how these steps read in the formal text.

Theorem 7.4.7: 2-cocycles and gerbes with local objects

Definitions. An A-gerbe is a type G that is a 1-type, merely inhabited and connected ($\|x = y\|$ for all x, y), with for each x a group isomorphism $\beta_x : (x = x) \to A$, the band. Being a 1-type follows from the band in [1]; it is included as a field, as in the note *About torsors and gerbes and liens*, and is a proposition, so the type of gerbes is the same. The compatibility of the bands with conjugation, $\beta_y(q^{-1}\ell q) = \beta_x(\ell)$ for $q : x = y$, is not part of the definition: it holds for any function of x, by path induction on q. An L-2-cocycle is $c : L^3 \to A$ with $c_{ijk} + c_{ikm} = c_{jkm} + c_{ijm}$, which is the condition of [1] rearranged. The statement is that

$$\textstyle\sum_{G:A\text{-gerbe}} \sum_{u:L\to G} \prod_{i,j}(u_i = u_j) \;\longrightarrow\; L\text{-2-cocycles}, \qquad c_{ijk} = \beta_{u_i}(p_{ij}\, p_{jk}\, p_{ik}^{-1}),$$

is an equivalence, for $\|L\|$. The gerbes are in the same universe as L and A. The paper gives no proof, only a reference to Deligne's construction [2, 5.3].

Torsors. Paths between two torsors are the same as equivariant maps: the type of pairs (Y, f) with $f : X \to Y$ equivariant is contractible, since an equivariant map of torsors is a bijection and by induction on equivalences. Equivariant maps $X \to X$ are the translations, so $\mathrm{Aut}(X) \simeq A$. Hence the type of torsors is a gerbe, the trivial gerbe. It lies one universe higher, but its paths are small. The image K of the single torsor A, given by replacement, is small, and contains every torsor since all torsors are merely isomorphic. K is the small trivial gerbe.

Paths in a gerbe. In a gerbe the paths $x = y$ form an A-torsor, with $a \cdot h = \beta_x^{-1}(a)\, h$. A map of gerbes which preserves the bands is therefore an equivalence: on paths it is an equivariant map of torsors, hence a bijection, so the map is an embedding; and it is surjective because the target is connected.

The cocycle. c is a 2-cocycle: both sides of the cocycle law are $\beta_{u_i}(p_{ij}p_{jk}p_{km}p_{im}^{-1})$, after moving c_{jkm} from u_j to u_i by conjugation with p_{ij}.

The proof. Being an equivalence is a proposition, so we may assume a point $i_0 : L$. We show that each fibre is contractible.

- *Centre.* Take K, with all points equal to the base point and as path from i to j the loop of band c_{i_0ij}. Its cocycle is $c_{i_0ij} + c_{i_0jk} - c_{i_0ik}$, which is c_{ijk} by the cocycle law at (i_0, i, j, k).
- *Contraction.* Given (G, u, p) with cocycle c, map $y : G$ to the torsor of paths $u_{i_0} = y$, viewed in K. This map preserves the bands: a loop ℓ at y acts on paths by composition, which is the translation by $\beta_y(\ell)$. So it is an equivalence. It sends u_i to the base point, using $h \mapsto \beta(h\, p_{i_0i}^{-1})$, and p_{ij} to the loop of band c_{i_0ij}. A structure identity principle, by induction on equivalences, turns this into an equality with the centre.

Comparison with Deligne. Deligne's inverse is the gerbe of families of torsors P_i with isomorphisms p_{ij} satisfying the cocycle condition twisted by c. The formal proof does not need it: with a point of L, which may be assumed, every gerbe with points is identified with the trivial one, and the comparison map is Deligne's map $P \mapsto \mathrm{Hom}(Q_i, P)$ at the single index i_0. The proof is about 1400 lines; most of it is the path algebra of the trivialisation.

Theorem 7.4.8: Čech H^2 of a 2-acyclic cover

Changing the chosen paths. Let (G, u, p) be a gerbe with points and paths, with cocycle c, and b a 1-cochain. Replacing each p_{ij} by $b_{ij} \cdot p_{ij}$ (composition with the loop of band b_{ij}) keeps the gerbe and the points, and the new cocycle is

$$c_{ijk} + (\delta b)_{ijk}, \qquad (\delta b)_{ijk} = b_{ij} + b_{jk} - b_{ik}.$$

The proof takes place in the torsors of paths: c_{ijk} is the difference from p_{ik} to $p_{ij}p_{jk}$, and prefixing a path is equivariant ($q\,(a \cdot h) = a \cdot (q\, h)$, by the compatibility of the band with conjugation). Hence $(b_{ij}p_{ij})(b_{jk}p_{jk}) = (b_{ij} + b_{jk} + c_{ijk}) \cdot p_{ik}$, and differences in a torsor are unique.

Objects with paths are coboundaries. For $\|L\|$,

$$\textstyle\sum_{y:G} \prod_i (u_i = y) \;\simeq\; \sum_b (c + \delta b = 0),$$

by $(y, r) \mapsto b$, b_{ij} the difference from p_{ij} to $r_i\, r_j^{-1}$. The paths r_i are part of the data on the left, so nothing is chosen; a point of L is assumed only to prove that the map is an equivalence, a proposition. So c is a coboundary exactly when merely there is an object with paths to all the u_i.

Setting. X is a type with a finite cover $U_1, \ldots, U_n$ (every x is merely in some U_i) and F a family of abelian groups over X. With $L_x = \sum_i U_i(x)$, a Čech 2-cocycle is an L_x-2-cocycle for each x, as in the paper's proof of Theorem 7.4.5. For a cocycle c, G_c is the family of gerbes given at each x by the inverse of Theorem 7.4.7. Two cocycles are cohomologous when they differ by δb; this is an equivalence relation.

Acyclicity. The cover is 2-acyclic when $H^1(U_i \cap U_j, F) = 0$ for all i, j (including $i = j$) and $H^2(U_i, F) = 0$ for all i, that is, every family of torsors over $U_i \cap U_j$ is merely equal to the trivial family, and every family of gerbes over U_i merely equal to the trivial family. This is equivalent to: such families merely have a section. Pointwise, a torsor has an element if and only if it is equal to A, and a gerbe has an object if and only if it is equal to the trivial gerbe (the trivialisation of Theorem 7.4.7 with one point); for families both directions are pointwise, by function extensionality.

Finite choice. Acyclicity gives, for each i, merely a section over the whole U_i, and for each i, j merely a section over the whole $U_i \cap U_j$. Since the cover is finite, these finitely many mere sections are combined into one by finite choice,

$$\textstyle\prod_{i:\mathrm{Fin}\, n} \|P_i\| \to \|\prod_{i:\mathrm{Fin}\, n} P_i\|,$$

proved by recursion on n (at higher type). No choice over the points of X is needed: the sections are given on whole opens by acyclicity.

The theorem. Let $\check{H}^2$ be the set of cocycles modulo being merely cohomologous, and

$$H^2(X, F) = \|\textstyle\prod_x F_x\text{-gerbe}\|_0,$$

the families of gerbes modulo mere equality; both are effective quotients constructed from replacement. The map $c \mapsto [G_c]$ induces $\check{H}^2 \simeq H^2(X, F)$, an equivalence of sets:

- *well defined*: cohomologous cocycles give the same family, by the change of paths and Theorem 7.4.7;
- *injective*: if G_c and $G_{c'}$ are merely equal, carry the points and paths of c' to the gerbes of c; acyclicity on the U_i gives paths between the two systems of points, conjugating by them does not change the cocycle, and the change of paths gives $c' = c + \delta b$;
- *surjective*: for a family of gerbes, acyclicity gives local objects on the U_i and paths on the $U_i \cap U_j$, and Theorem 7.4.7 gives the cocycle.

No separate gluing lemma for 1-types is needed: Theorem 7.4.7 already constructs the gerbe of a cocycle at each point.

Addition. The paper states an equality of groups. Addition on H^2 is defined from the sum of gerbes, constructed in three steps.

- *Sum of torsors.* For A-torsors T, S, the pairs (Z, φ) with $\varphi : T \times S \to Z$ satisfying $\varphi(a \cdot t, s) = a \cdot \varphi(t, s) = \varphi(t, a \cdot s)$ form a contractible type (assume points t_0, s_0; the centre is A with $(t - t_0) + (s - s_0)$). Its centre is $T \otimes S$ with $q : T \times S \to T \otimes S$, $q(a \cdot t, b \cdot s) = (a + b) \cdot q(t, s)$, and $\mathrm{Hom}(T \otimes S, Z)$ is equivalent to the bilinear maps.

- *Sum of gerbes.* For gerbes G_1, G_2 and $H = G_1 \times G_2$, let $F_g(h) = (h_1 = g_1) \otimes (h_2 = g_2)$, a family of torsors on H (loops at the source act, so moving the target along a loop ℓ is the translation by $+\beta(\ell)$). $G_1 + G_2$ is the image of $g \mapsto F_g$, small by replacement, with $\mu : G_1 \times G_2 \to G_1 + G_2$. A loop of F_g is a function $H \to A$; since H is connected and inhabited, $c \mapsto (\lambda h.c)$ is an equivalence $A \simeq (H \to A)$, and the band is the constant value, with no point chosen. Then $\beta(\mu(\ell_1, \ell_2)) = \beta(\ell_1) + \beta(\ell_2)$.
- *Cocycles.* For systems (u, p) and (u', p') with cocycles c, d, the points $\mu(u_i, u'_i)$ and paths $\mu(p_{ij}, p'_{ij})$ have cocycle $c + d$. Hence the family of gerbes of $c + d$ is equal to the sum of those of c and d.

The zero cocycle is realised by the trivial gerbe with constant points and identity paths, so the zero of H^2 is the class of the trivial family. The group laws of $\check{H}^2$ hold pointwise on cocycles, and are transferred to H^2 through the equivalence, which preserves zero and addition. So $\check{H}^2 \simeq H^2(X, F)$ as abelian groups.

Refinement. If a finite cover $V_1, \ldots, V_{n'}$ refines $U_1, \ldots, U_n$ through ρ with $V_j \subseteq U_{\rho(j)}$, cocycles pull back along $(j, v) \mapsto (\rho(j), v)$. Restricting the points and paths of a gerbe pulls back its cocycle, so the family of gerbes of the pulled-back cocycle is the same family. Hence, for 2-acyclic covers, the refinement map on Čech classes commutes with the two comparison maps to $H^2(X, F)$; it is therefore an additive isomorphism, independent of ρ, and refinement satisfies the identity and composition laws (by naturality and injectivity of the comparison maps). Any two 2-acyclic covers are compared canonically through $H^2(X, F)$. A common refinement receives the two Čech refinement maps whether or not it is acyclic (the pull-back of cocycles and of coboundaries uses no acyclicity); it is 2-acyclicity of the refinement that makes its own comparison map an isomorphism, and so lets the comparison of the two covers be read through it. In the formal text the quotient $\check{H}^2$ is built inside the segment that assumes acyclicity, so the refinement map on classes is stated there for 2-acyclic covers.

4 Summary

directory	content	hypotheses	lines	time
`replacement`	truncation, quotients	—	165	0.02 s
`sag`	Lemma 2.2.4	$1 \neq 0$, Nullst. for $R[X]/(f)$	807	0.03 s
`torsor`	7.4.2, 7.4.7, 7.4.8	—	3869	0.44 s
`bounded`	Theorem 3.3.3	$1 \neq 0$, two halves of SQC	428	0.07 s
`projective`	Ex. 6.1.9, Lemma 6.2.1	Loc, SQC instances	1141	0.11 s
`polynomial`	$A[X]$, universal property	—	923	0.11 s
`presentations`	presentations, Prop. 2.2.2, Thm. 3.3.3, Ex. 6.1.9 (algebra)	Loc, uniform SQC	1473	0.40 s
`sag-uniform`	Lemma 2.2.4	Loc, uniform SQC	439	0.32 s

Lines count the files of each directory, not the files it shares with others by links; times are for checking everything a directory imports. `sag-uniform` combines the proof of `sag` with the Nullstellensatz of `presentations`.

What is not formalised: localisation, quotients by elements and tensor products as operations on presentations; Z-choice and the results that need it; and the interpretation of the axioms in a model.

A Excerpts of the formal proof of Lemma 7.4.2

The file is `sag/torsor/Torsor.seg`, about 340 lines. The excerpts below are copied from it. Only `Laws` and `Coc` are shortened (the file names the intermediate Σ-types), and `...` marks an omitted proof. In `Unit`, `(x : A) * B` is a Σ-type, `w.1`, `w.2` are its projections, `Id` is the identity type, and `calc A { a =< p > b =< q > c }` is an equational chain. `Tr.Trunc[X].carrier` is the truncation $\|X\|$ obtained from replacement, and `univ` is the univalence parameter.

Torsors, cocycles and differences. A torsor structure is the action with its laws. Free and transitive is the contractibility of $\sum_a a \cdot t = t'$. The difference $t' - t$ (`diff`) is the centre of contraction, and `uniq` says that it is the only solution.

```
AD (T : U l) (act : A -> T -> T) : U l =
  (a b : A) (t : T) -> Id T (act (add a b) t) (act a (act b t)) ;
ZE (T : U l) (act : A -> T -> T) : U l = (t : T) -> Id T (act o t) t ;
FT (T : U l) (act : A -> T -> T) : U l =
  (t t' : T) -> isContr ((a : A) * Id T (act a t) t') ;
Laws (T : U l) (act : A -> T -> T) : U l = (sT : isSet T) * Rest T act ;
                                    -- Rest: AD, ZE, FT and ||T||
TorStr (T : U l) : U l = (act : A -> T -> T) * Laws T act ;
Left : U l+ = (T : U l) * (s : TorStr T) * (L -> T) ;
Coc : U l = (c : L -> L -> A) * ((i j k : L) -> Id A (add (c i j) (c j k)) (c i k)) ;

diff (t t' : T) : A = (ft t t').1.1 ;
dact (t t' : T) : Id T (act (diff t t') t) t' = (ft t t').1.2 ;
uniq (t t' : T) (a : A) (p : Id T (act a t) t') : Id A (diff t t') a =
  ap ((b : A) * Id T (act b t) t') A (\ w -> w.1) (ft t t').1 (a, p) ((ft t t').2 (a, p)) ;
```

From torsors to cocycles. The cocycle law is `uniq` applied to the chain $(c_{ij} + c_{jk}) \cdot u_k = c_{ij} \cdot (c_{jk} \cdot u_k) = c_{ij} \cdot u_j = u_i$ (here `w = (T, s, u)`).

```
c (i j : L) : A = diff (w.2.2 j) (w.2.2 i) ;
coc (i j k : L) : Id A (add (c i j) (c j k)) (c i k) =
  symH (uniq (w.2.2 k) (w.2.2 i) (add (c i j) (c j k)) (calc w.1 {
    act (add (c i j) (c j k)) (w.2.2 k)
      =< ad (c i j) (c j k) (w.2.2 k) > act (c i j) (act (c j k) (w.2.2 k))
      =< ap w.1 w.1 (act (c i j)) (act (c j k) (w.2.2 k)) (w.2.2 j) (dact (w.2.2 k) (w.2.2 j))
      > act (c i j) (w.2.2 j)
      =< dact (w.2.2 j) (w.2.2 i) > w.2.2 i })) ;
```

From cocycles to torsors. T_c, its action and its points:

```
Cond (v : L -> A) : U l = (k m : L) -> Id A (v m) (add (v k) (c.1 k m)) ;
Tc : U l = (v : L -> A) * Cond v ;
actc (a : A) (v : Tc) : Tc =
  (\ k -> add a (v.1 k),
   \ k m -> transH (ap A A (add a) (v.1 m) (add (v.1 k) (c.1 k m)) (v.2 k m))
                   (symH (assoc a (v.1 k) (c.1 k m)))) ;
uc (i : L) : Tc = (\ k -> c.1 i k, \ k m -> symH (c.2 i k m)) ;
```

Free and transitive is proved by `ftk k0 v w`, given a point `k0`, and the point is obtained from $\|L\|$ (`hL`) because the goal is a proposition:

```
ft-c : FT Tc actc = \ v w ->
  Tr.Trunc[L].induction @l (\ x -> isContr ((a : A) * Id Tc (actc a v) w))
    (\ x -> isProp-isContr ((a : A) * Id Tc (actc a v) w)) (\ k0 -> ftk k0 v w) hL ;
```

The map e. $e(t) = (k \mapsto t - u_k)$, with the proof that it lies in T_c. The proof that e is a bijection (`fib`) uses a point of L in the same way as `ft-c`.

```
e (t : T) : PsiM[phi w].Tc =
  (\ k -> diff (u k) t,
   \ k m -> uniq (u m) t (add (diff (u k) t) (c k m)) (calc T {
     act (add (diff (u k) t) (c k m)) (u m)
       =< ad (diff (u k) t) (c k m) (u m) > act (diff (u k) t) (act (c k m) (u m))
       =< ap T T (act (diff (u k) t)) (act (c k m) (u m)) (u k) (dact (u m) (u k))
       > act (diff (u k) t) (u k)
       =< dact (u k) t > t })) ;
eqv : Equiv T PsiM[phi w].Tc =
  (e, qinv T PsiM[phi w].Tc e (\ y -> (fib y).1.1) (\ y -> (fib y).1.2) ...) ;
path : Id Left w (psi (phi w)) =
  sip w.1 w.2.1 w.2.2 PsiM[phi w].Tc eqv PsiM[phi w].str PsiM[phi w].uc (e-act, e-pt) ;
```

Univalence. Induction on equivalences, from univalence, and its use: an equivariant equivalence preserving the points gives an identification in `Left`. It is enough to prove this for the identity (`sip-base`), by function extensionality on the action and propositionality of the laws.

```
eqind (T : U l) (P : (T' : U l) -> Equiv T T' -> U l+) (d : P T (idEquiv T))
  (T' : U l) (e : Equiv T T') : P T' e =
  Prelude[l+].subst (Equiv T T') (P T')
    (Axioms[l].idtoeqv T T' ((univ @l T T').sec e)) e ((univ @l T T').sec-eq e)
    (idJ (U l) T (\ X q -> P X (Axioms[l].idtoeqv T X q)) d T' ((univ @l T T').sec e)) ;

SipMot (T : U l) (s : TorStr T) (u : L -> T) (T' : U l) (e : Equiv T T') : U l+ =
  (s' : TorStr T') (u' : L -> T') -> Eqv T T' s s' u u' e
  -> Id Left (T, s, u) (T', s', u') ;
sip (T : U l) (s : TorStr T) (u : L -> T) (T' : U l) (e : Equiv T T')
  : SipMot T s u T' e = eqind T (SipMot T s u) (sip-base T s u) T' e ;
```

The lemma.

```
lemma-7-4-2 : Prelude[l+].Equiv Left Coc =
  (phi, Prelude[l+].qinv Left Coc phi psi phipsi psiphi) ;
```

References

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[2] P. Deligne. *Le symbole modéré.* Publ. Math. IHÉS 73, 147–181, 1991.

[3] E. Cavallo, T. Coquand. *Type-theoretic replacement and univalent completion: applications and interpretations.* TYPES 2025, LIPIcs.